

EDUCATION

University of California San Diego

Ph.D. Student, Structural Engineering

San Diego, CA

September 2025 – Present

- Coursework: Mechanical Design of Composite Structures, Finite Element Analysis, Nondestructive Evaluation.

University of Wisconsin–Madison

Bachelor of Science, Mechanical Engineering

Madison, WI

May 2022

ENGINEERING EXPERIENCE

University of California San Diego, Tehrani Group

Graduate Student Researcher

San Diego, CA

September 2025 – Present

- Thermoplastic composites, AFP manufacturing and troubleshooting, structural testing, NDE, microscopy, failure analysis, and experimental process development.

Collins Aerospace

Manufacturing Engineer II

Jacksonville, FL

August 2023 – February 2025

- Responsible manufacturing engineer for low-rate composite aerospace hardware; manufacturing development, tooling, bonding, machining, technician support, and production troubleshooting.

Research Engineer

July 2022 – August 2023

- Prototype composite process and material-system development, tooling, supplier coordination, production drawings, NDE calibration hardware, fabrication, machining, and inspection.

Mechanical Engineering Intern

May 2021 – August 2021

- Production troubleshooting, tooling design and fabrication, machining, welding, and root-cause investigation.

Pierce Manufacturing

Manufacturing Engineering Co-op

Appleton, WI

June 2020 – December 2020

- Manufacturing tooling, robotic-welding DFM, engineering drawings, safety improvements, and technician production support.

SELECTED DESIGN, BUILD & TEST PROJECTS

Composite Aerospace Structure Manufacturing Development

Manufacturing Process Design, First-Article Build & Production Ownership

Collins Aerospace

2023 – 2025

- Owned manufacturing development and serial production of a complex composite aerospace structure, defining and refining layup methods, cure processes, surface preparation, bonding, machining integration, handling, tooling, and work instructions.
- Worked directly with technicians to troubleshoot first-build issues, improve tooling and ergonomics, simplify difficult manufacturing features, and carry lessons learned into subsequent units to improve manufacturability.

High-Viscosity E-Glass Infusion Process

Composite Process Development, Tooling Design & Prototype Manufacturing

Collins Aerospace

2022 – 2023

- Developed a vacuum-infusion process for an exceptionally thick E-glass laminate using a highly viscous, fast-curing urethane system, replacing conventional edge-to-edge infusion with through-thickness flow to minimize resin travel distance.
- Designed and fabricated a large-bore, center-out distribution manifold to reduce viscous pressure losses and control the resin front, successfully producing a fully infused prototype on the first full attempt and generating customer interest in further development.

AFP Wedge-Gap Defect Research

Experimental Design, Robotic Manufacturing, Structural Test & Failure Analysis

UC San Diego

2025 – Present

- Design and manufacture controlled CF/LM-PAEK laminates containing intentional automated-fiber-placement gaps and overlaps to isolate the structural effects of manufacturing-induced defects.
- Carry specimens through the complete experimental cycle from AFP programming and fabrication through post-processing, ultrasonic C-scan inspection, specimen preparation, mechanical testing, microscopy, fractography, data interpretation, and test-article redesign.
- Evaluate defect effects through Mode I and Mode II fracture, and compression testing while using cross-sectional microscopy and fracture-surface characterization to connect local laminate geometry with structural behavior.
- Iterate laminate and specimen designs when results reveal competing effects or unexpected behavior, including investigation of visible fiber waviness that does not produce the anticipated reduction in compressive performance.

Ultrasonic A-Scan Calibration Block Development

Collins Aerospace

Mechanical Design, Composite Fabrication, Machining & NDE Validation

2022 – 2023

- Designed a compact ultrasonic calibration standard reproducing critical thicknesses, angled back-wall geometry, machined features, and intentional defect conditions from a substantially larger production composite structure.
- Developed the manufacturing approach and work instructions, fabricated the composite article, and worked directly with machinists to create complex final geometry while preserving features required for inspection.
- Iterated the design and manufacturing process through multiple builds, then worked with NDE technicians to validate the calibration article against production hardware during ultrasonic A-scan inspection.

Baja SAE Lower Control Arm Redesign

Wisconsin Baja SAE

Vehicle Testing, Load Development, FEA, Design, Fabrication & Validation

2019 – 2022

- Instrumented the Baja vehicle with strain gauges and accelerometers and collected data over representative obstacles to estimate real suspension loads for structural analysis.
- Used measured loads and iterative SolidWorks FEA to redesign the lower control arm as a lightweight welded sheet-metal box structure, balancing strength, buckling resistance, manufacturability, impact tolerance, and mass.
- Reduced weight through iterative geometry changes and material removal, then fabricated and validated the final design through vehicle testing and competition use, learning to treat simulation as one design input rather than an absolute answer.

Baja SAE Four-Wheel-Drive System Development

Wisconsin Baja SAE

Drivetrain Architecture, Fabrication, Testing & Failure Analysis

2021 – 2022

- Helped develop the team's first four-wheel-drive Baja vehicle, evaluating drivetrain architectures against competing requirements for mass, debris tolerance, packaging, manufacturability, and durability.
- Selected and developed a lightweight belt-driven concept in place of heavier driveshaft and debris-sensitive chain alternatives, iterating pulley layouts, belt paths, mounting hardware, and supporting structures.
- Fabricated and assembled the system using machined components, welded structures, and custom fixtures while resolving alignment, packaging, installation, and manufacturing issues during vehicle integration.
- Repeated testing exposed belt slip and breakage during severe transient wheel-impact events, demonstrating the importance of identifying dynamic load cases and component failure modes early in system architecture rather than optimizing primarily for nominal operation.

Morgan 34 Sailboat Restoration & Steering Conversion

Independent Project

Composite Structural Repair, Wet Layup & Mechanical Systems Integration

- Restored a 34-ft fiberglass sailboat purchased from a Hurricane Sandy salvage auction after severe hull impact damage, removing cracked laminate and preparing a large irregular structural repair area.
- Rebuilt the damaged hull using a tapered fiberglass scarf repair and wet layup techniques, gaining early hands-on experience with damage removal, surface preparation, epoxy, fillers, reinforcement, and composite repair.
- Converted the vessel from tiller to wheel steering by integrating a custom pulley and cable system, glassing fabricated mounts and mechanical hardware into the existing composite structure in confined interior spaces.

2003 Chevrolet Tahoe Maintenance & Repair

Independent Project

Mechanical and Electrical Troubleshooting

Ongoing

- Independently diagnose and repair mechanical and electrical systems on a 250,000-mile Chevrolet Tahoe, including suspension, drivetrain, brakes, charging system, and instrument-cluster electronics.

TECHNICAL SKILLS

Mechanical Design & Analysis: Finite Element Analysis (FEA), SolidWorks, CATIA, GD&T, engineering drawings, composite laminate design, root-cause analysis, experimental design

Manufacturing & Fabrication: CNC milling, CNC turning, waterjet cutting, MIG welding, tooling, fixtures and jigs, prototype manufacturing, composite machining

Composites: Automated Fiber Placement (AFP), prepreg layup, resin infusion, out-of-autoclave processing, wet layup, bonding, co-bonding, secondary bonding, composite repair

Test & Characterization: Mechanical testing, ultrasonic A-scan and C-scan inspection, microscopy and metrology, dimensional inspection, fractography, defect and failure characterization, environmental conditioning

Robotics & Software: KUKA robotic manufacturing systems, AddPath, MATLAB, Mathematica, L^AT_EX

ADDITIONAL

Security Clearance: Previously held U.S. Secret security clearance while employed at Collins Aerospace

Research & Presentations: Three manuscripts in preparation; oral presenter at the American Society for Composites Annual Technical Conference; presented research at an Office of Naval Research (ONR) program review

Leadership: Steering & Suspension Kinematics Team Lead, Wisconsin Baja SAE; Founding Member & Secretary, UC San Diego SAMPE Student Chapter